

- 8 The map shows six islands near the coast of California.



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Each of these islands has a population of island foxes descended from the California grey fox.

Each island fox population has developed different characteristics from the fox populations on the other islands.

The photograph shows a fox from the population found only on the island of Santa Cruz.



(Source: © Christian Loader/Alamy Stock Photo)

- (a) (i) Which of the following is a term used to describe the fact that this subspecies of island fox is found only on Santa Cruz?

(1)

- ☐ A diversity
- ☐ B endemic
- ☐ C mutation
- ☐ D specific



- (ii) Explain how isolation caused the formation of six fox populations with different characteristics.

Use the information in the question to support your answer.

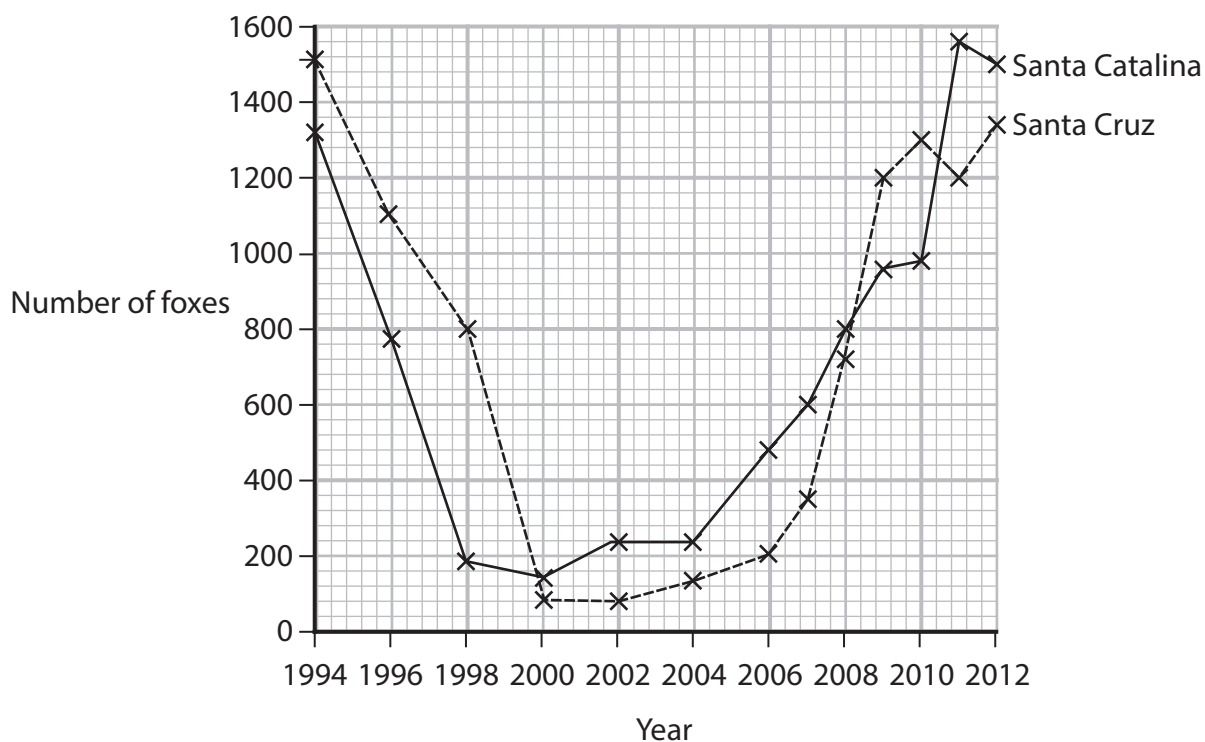
(4)

- (iii) Explain how molecular evidence could be used to determine which of the other five fox populations is the most closely related to the fox population on Santa Cruz.

(2)



(b) The graph shows the sizes of two fox populations from 1994 to 2012.



Two main causes for the decrease in island fox populations from 1994 to 2000 were predation by golden eagles and diseases transmitted from dogs and raccoons.

- (i) Calculate the percentage decrease in the Santa Cruz fox population from 1994 to 2002.

(1)

Answer%

- (ii) Explain how the Hardy–Weinberg equation can be used to show the change in the frequency of a recessive allele in the Santa Cruz population from 1994 to 2000.

(2)

.....

.....

.....

.....

.....

.....



(iii) In 2004, a conservation plan was produced to:

- increase the sizes of the Santa Cruz and Santa Catalina fox populations
- maintain the genetic diversity of the Santa Cruz and Santa Catalina fox populations.

Explain how conservation could achieve these goals.

Use the information in the question and your own knowledge to support your answer.

(4)

(Total for Question 8 = 14 marks)

TOTAL FOR PAPER = 80 MARKS

